

An Internship Report

Of

Network Technician Career Path

Submitted

To

**School of Computer Science and Engineering
IILM University, Greater Noida, U.P.**

In partial fulfilment of the requirement of degree of
B.Tech (Computer Science and Engineering)



By

Roll Number of Student: 2410031474

Name of Student: ANKIT KUMAR

Batch: 2024-2028

Candidate's Declaration

I, ANKIT KUMAR, do hereby declare that the internship report titled “Network Technician Career Path” has been completed by me to fulfil the requirement for the award of the degree of Bachelor of Technology in Computer Science and Engineering. All the references have been quoted and I have not taken as such any material from any other source. I affirm that this report is my own and has not been submitted to any other institute for any degree/diploma requirement, and further I shall be solely responsible for any kind of copyright violation in this regard.

Signature:-

Student Name: ANKIT KUMAR

Roll No.: 2410031474

Acknowledgement

I am using this opportunity to express my gratitude to everyone who supported me throughout the internship. I am thankful for their aspiring guidance, invaluable constructive criticism, and friendly advice during this work.

I would like to thank the Cisco Networking Academy program and IILM University for providing me the opportunity and the structured platform to learn and gain hands-on exposure in the domain of computer networking through the Network Technician Career Path. I am also thankful to my institute, IILM University, Greater Noida, and the faculty of the School of Computer Science and Engineering for their continuous support and encouragement.

I express my gratitude to my parents for their blessings.

Signature:-

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Table of Contents

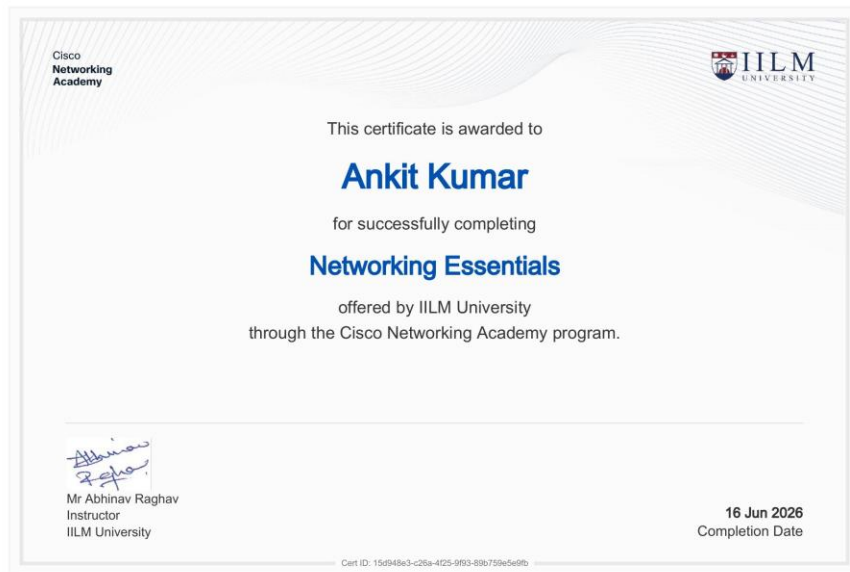
1. Candidate's Declaration	i
2. Acknowledgement	ii
3. Internship Certificates of Completion	-
4. Project Description	
4.1 Introduction	1
4.2 Organization Profile	2
4.3 Problem Statement	3
4.4 Project Objectives	3
4.5 Scope of the Project	4
4.6 Technologies and Tools Used	4
4.7 System Architecture	5
4.8 Methodology	6
4.9 Expected Outcomes	7
4.10 Certificates and Communication Proof	8
5. Bibliography/References	9

3. Internship Certificates of Completion

The following three certificates, issued through the Cisco Networking Academy program in collaboration with IILM University, confirm successful completion of the Network Technician Career Path program referenced in this report.



Certificate: Network Technician Career Path — Cisco-Verified Credential — Issued: 16 Jun 2026



Certificate: Networking Essentials — Completion Date: 16 Jun 2026 — Cert ID: 15d948e3-c26a-4f25-9f93-89b759e5e9fb



Certificate: Exploring Networking with Cisco Packet Tracer — Completion Date: 16 Jun 2026 — Cert ID: 85092b01-6cb0-4f41-ab43-b8ebb5309f7f

4. Project Description

4.1 Introduction

This report presents a detailed account of the Network Technician Career Path undertaken through the Cisco Networking Academy, in collaboration with the School of Computer Science and Engineering, IILM University, Greater Noida. The program was carried out between April 2026 and June 2026 as part of the requirement of the Bachelor of Technology (Computer Science and Engineering) programme. The primary aim of this internship was to bridge the gap between the theoretical understanding of computer networking taught in the classroom and the practical, industry-aligned skills required of an entry-level network technician.

The program followed a structured curriculum comprising Networking Essentials, Exploring Networking with Cisco Packet Tracer, and the capstone Network Technician Career Path course, combining conceptual learning with hands-on simulation exercises and assessments. On successful completion of all requirements, a Cisco-verified Network Technician Career Path credential was awarded, along with course-completion certificates issued jointly by IILM University and Cisco Networking Academy.

4.2 Organization Profile

Cisco Networking Academy is Cisco's global IT skills and career-building program, delivered in partnership with academic institutions worldwide, including IILM University. It offers a structured, industry-recognized curriculum covering networking fundamentals, cybersecurity, and related IT disciplines, combining self-paced online learning with instructor-led sessions and simulation-based labs using Cisco Packet Tracer. Through its collaboration with the School of Computer Science and Engineering, IILM University enables students to earn Cisco-verified credentials as part of their academic and career preparation.

4.3 Problem Statement

Many engineering students study computer networks as a theoretical subject but have limited exposure to configuring, troubleshooting, and securing real or simulated network devices such as routers, switches, and wireless access points. Understanding protocols and addressing schemes on paper is only one part of network literacy; being able to diagnose connectivity issues, configure secure access, and apply help desk best practices is an equally important, and often under-practiced, skill. The problem addressed by this internship was to gain and demonstrate practical competence across the full scope of entry-level network technician tasks, from standards and addressing through to troubleshooting and secure user access.

4.4 Project Objectives

- To explain the importance of standards and protocols in network communications.
- To explain how communication occurs on Ethernet networks.
- To explain the features of IPv4 and IPv6 addresses.
- To configure an integrated wireless router and wireless client to connect securely to the internet.
- To explain how Ethernet operates in a switched network.
- To compare the operations of transport layer protocols in supporting end-to-end communication.
- To explain how DNS and DHCP services operate and diagnose basic network connectivity issues.
- To explain how routers use network layer protocols and services to enable end-to-end connectivity.
- To demonstrate effective troubleshooting methodologies and help desk best practices.
- To configure secure user access on a network.

4.5 Scope of the Project

The scope of this internship spans foundational to intermediate computer networking concepts — network standards, Ethernet switching, IPv4/IPv6 addressing, wireless configuration, transport and network layer protocols, DNS/DHCP services, routing, and troubleshooting. Practical exercises were carried out in a simulated environment using Cisco Packet Tracer, and did not extend to configuration of live enterprise production networks. The program is designed to prepare a student for entry-level network technician and help-desk roles.

4.6 Technologies and Tools Used

Technology / Tool	Purpose
Cisco Packet Tracer	Simulation environment used to build, configure, and test networks

Technology / Tool	Purpose
IPv4 and IPv6	Addressing schemes studied and applied across exercises
Switching and Ethernet	Switched network operation and frame forwarding
Wireless Networking	Integrated wireless router and client configuration
Transport & Network Layer Protocols	TCP/UDP concepts and routing fundamentals
Network Services (DNS, DHCP)	Name resolution and automatic address assignment
Security & Support	Secure user access configuration and help desk troubleshooting methodology

4.7 System Architecture

The typical small-to-medium network architecture studied and simulated during the program can be summarized in the following layers: an End-Device Layer of client PCs, laptops and wireless devices; an Access Layer of switches and a wireless router/access point providing Ethernet and secure wireless connectivity; a Network Layer of routers using network layer protocols and routing services for end-to-end connectivity; a Services Layer of DNS and DHCP servers providing name resolution and automatic address assignment; a Transport Layer governing reliable and best-effort end-to-end communication; and a Management & Support Layer of secure user access controls and structured troubleshooting/help-desk procedures.

4.8 Methodology

The internship followed a structured, course-wise methodology combining conceptual learning, simulation-based labs, and assessments, as summarized in the table below.

Module	Description	Completion Date
Networking Essentials	Foundational concepts: network standards and protocols, Ethernet communication, IPv4/IPv6 addressing, and basic network media	16 Jun 2026
Exploring Networking with Cisco Packet Tracer	Hands-on simulation using Cisco Packet Tracer to build, configure and test small network topologies, reinforcing theoretical concepts through practice	16 Jun 2026
Network Technician Career Path (Capstone)	Integrated, career-oriented path covering switched network operation, wireless router/client configuration, transport-layer protocols, DNS/DHCP services, routing, troubleshooting methodology and secure user access — culminating in the Cisco-verified credential	16 Jun 2026

Each course concluded with assessments to evaluate conceptual understanding, and the capstone Network Technician Career Path course required demonstration of practical skills across all the topics covered. On successful completion of the final assessment, the Cisco-verified Network Technician Career Path credential was issued.

4.9 Expected Outcomes

- Ability to explain and apply key networking standards and protocols.
- Practical understanding of Ethernet switching and IPv4/IPv6 addressing.
- Hands-on capability to configure a wireless router and client for secure internet access.
- Understanding of transport and network layer protocols supporting end-to-end communication.
- Ability to diagnose basic connectivity issues using DNS and DHCP knowledge.
- Competence in structured troubleshooting methodology and help desk best practices.
- Ability to configure secure user access on a network.
- A Cisco-verified, industry-recognized Network Technician Career Path credential.

4.10 Certificates and Communication Proof

The Network Technician Career Path verified credential, along with the Networking Essentials and Exploring Networking with Cisco Packet Tracer course-completion certificates issued jointly by IILM University and Cisco Networking Academy, confirming successful completion of the program, are attached in Section 3 of this report as proof of successful completion.

5. Bibliography/References

1. Cisco Networking Academy, “Network Technician Career Path,” Verified Course Completion, Issued on 16 Jun 2026.
2. IILM University / Cisco Networking Academy, “Networking Essentials”, Certificate of Completion, Cert ID: 15d948e3-c26a-4f25-9f93-89b759e5e9fb, 2026.
3. IILM University / Cisco Networking Academy, “Exploring Networking with Cisco Packet Tracer”, Certificate of Completion, Cert ID: 85092b01-6cb0-4f41-ab43-b8ebb5309f7f, 2026.
4. Cisco Networking Academy, <https://www.netacad.com/>
5. Cisco Packet Tracer Documentation, <https://www.netacad.com/courses/packet-tracer>
6. Cisco, “Understanding IPv4 and IPv6 Addressing,” Cisco Documentation.